

AWS EBS with Terraform with User Data

To Begin with the Lab

Summary of the Lab

This lab demonstrated how to automate AWS EC2 provisioning using Terraform with User Data and Cloud-Init. After creating the required Terraform configuration files and initialization scripts, Terraform was used to deploy an EC2 instance and automatically install Apache during launch. In the second setup, Cloud-Init was integrated using additional configuration files to automate instance customization. Finally, the Cloud-Init logs were verified through the `/var/log/cloud-init.log` file to confirm successful execution of the initialization process.

Prerequisite for the lab

Running EC2 instance with installed terraform

Now create a new directory and Navigate to the directory.

Create the files

`createinstance.tf`

`variable.tf`

`provider.tf`

`installapache.sh`

Then the run the command

`terraform init`

```
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume# terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.50.0...
- Installed hashicorp/aws v6.50.0 (signed by HashiCorp)

Initializing the backend...

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume#
```

Now run the command

```
ssh-keygen -f levelup_key
```

Hit enter

```

root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume# ssh-keygen -f levelup_key
Generating public/private ed25519 key pair.
Enter passphrase for "levelup_key" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in levelup_key
Your public key has been saved in levelup_key.pub
The key fingerprint is:
SHA256:+VLLMUwcVapL6RNSf/oBFyX+6pYjbJMcuDOPDXF0F0w root@ip-172-31-17-1
The key's randomart image is:
+--[ED25519 256]--+
|      .oo=Eo|
|      o.. +o|
|      B +..|
|      . = B o.|
|    S +.B...o|
|      o.*.oo+|
|      . o++ooo|
|      .++Bo+..|
|      . =o+...|
+-----[SHA256]-----+
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume#

```

Now run the command

terraform plan

it will ask you to enter the secret key

```

# aws_key_pair.levelup_key will be created
+ resource "aws_key_pair" "levelup_key" {
+   arn                = (known after apply)
+   fingerprint        = (known after apply)
+   id                 = (known after apply)
+   key_name            = "levelup_key"
+   key_name_prefix     = (known after apply)
+   key_pair_id         = (known after apply)
+   key_type            = (known after apply)
+   public_key          = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAINSUBzJ5BKf0p4b0YMN/ZBQV50nvHlxjDn+pCCVxuJou root@ip-172-31-17-1"
+   region              = "us-east-1"
+   tags_all            = (known after apply)
+ }

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ public_ip = (known after apply)

```

Use the command

terraform apply

```

aws_key_pair.levelup_key: Creating...
aws_key_pair.levelup_key: Creation complete after 1s [id=levelup_key]
aws_instance.MyFirstInstnace: Creating...
aws_instance.MyFirstInstnace: Still creating... [00m10s elapsed]
aws_instance.MyFirstInstnace: Still creating... [00m20s elapsed]
aws_instance.MyFirstInstnace: Still creating... [00m30s elapsed]
aws_instance.MyFirstInstnace: Creation complete after 32s [id=i-0e5aed9c8de7d6e4d]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

public_ip = "54.160.165.204"
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/UserData#

```

Now go to your browser and copy the ip address



Create a new directory

Create the files

createinstance.tf

variable.tf

provider.tf

installapache.sh

cloudinit.tf

init.cfg

Then the run the command

terraform init

```
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```

Hit enter

```
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume# ssh-keygen -f levelup_key
Generating public/private ed25519 key pair.
Enter passphrase for "levelup_key" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in levelup_key
Your public key has been saved in levelup_key.pub
The key fingerprint is:
SHA256:+VLlMUwcVapL6RNSf/oBFyX+6pYjbJMcuDOPDXF0F0w root@ip-172-31-17-1
The key's randomart image is:
+--[ED25519 256]--+
|      .oo=Eo|
|      o..+o|
|      B +..|
|      . = B o.|
|      S +.B...o|
|      o.*.oo+|
|      . o++ooo|
|      .++Bo+..|
|      .o+...|
+-----[SHA256]-----+
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume#
```

Now run the command

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```
Plan: 2 to add, 0 to change, 0 to destroy.  
Changes to Outputs:  
+ public_ip = (known after apply)
```

Now run the command

terraform apply

now connect to the machine

go to the log directory

cd /var/log

you will see the cloud-init.log file where we are storing all the log

```
ubuntu@ip-172-31-22-221:/var/log$ cd /var/log  
ubuntu@ip-172-31-22-221:/var/log$ ls  
README      amazon      apport.log  auth.log    chrony      cloud-init.log  dmesg      journal  landscape  private  sysstat      wtmp  
alternatives.log  apache2     apt         btmp        cloud-init-output.log  dist-upgrade  dpkg.log   kern.log  lastlog    syslog   unattended-upgrades
```

Use cat command to see the content of the file.